

INTRODUCTION



Algorithms are everywhere. You have probably executed a few already today. In this book, you will read about dozens of algorithms: some simple, some complex, some famous, some unknown, all interesting, and all worth learning. The first algorithm of the book is also the most delicious—it generates a berry granola parfait, and it’s shown in its entirety in [Figure 1](#). You may be accustomed to calling this type of algorithm a “recipe,” but it fits Donald Knuth’s definition of an *algorithm*: a finite set of rules that gives a sequence of operations for solving a specific type of problem.

Berry Granola Parfaits

Directions

1. Place one-sixth of one cup of blueberries at the bottom of a large serving glass.
2. Cover the blueberries with a half cup of plain Turkish yogurt.
3. Place one-third of one cup of granola on top of the yogurt.
4. Cover the granola with a half cup of plain Turkish yogurt.
5. Place strawberries on top of everything in the glass so far.
6. Top it all off with your favorite whipped cream.

[Figure 1](#): An algorithm: a finite set of rules that gives a sequence of operations for solving a specific type of problem

Parfait-making is not the only domain of life governed by algorithms. Every year, the US government requires each adult citizen to execute an algorithm, and strives to imprison those who fail to do so correctly. In 2017, millions of Americans fulfilled this duty by completing the algorithm shown in [Figure 2](#), which is taken from a form called 1040-EZ.